

$ab = ba \rightarrow \text{group}$
 $\frac{S \vee S'}{\mathbb{Z} \times \mathbb{Z}}$
 $S \vee S'$
 $\frac{S \vee S'}{\mathbb{Z} * \mathbb{Z}}$
 $T := \mathbb{Z} \times \mathbb{Z}$
 $T \hookrightarrow \mathbb{Z} * \mathbb{Z}$
 $f \hookrightarrow \mathbb{Z} * \mathbb{Z}$

[illegible]

Example:

S^1

$RP^2: \mathbb{Z}_2$
 $\langle a \mid a^2 = 1 \rangle$
 $a^2 = 1$
 $RP^2: \text{cell } 1, 2, 3$
 $\pi_1(RP^2) = \pi_1(\mathbb{R}P^2) = \mathbb{Z}_2$

Hand-drawn sketches on a grid background:

- Top left: A small circle with a horizontal line passing through its center, labeled $\mathbb{R}/\mathbb{Z}$ to its left.
- Top center: A small circle with a horizontal line passing through its center.
- Top right: A large circle containing a square, which in turn contains a smaller circle with an 'X' inside. To the right of this is a sketch of a function curve labeled f and S , with a vertical line and a small circle below it.
- Middle left: A small circle with a horizontal line passing through its center.
- Middle center: A small circle with a horizontal line passing through its center.
- Middle right: A large circle containing a square, which in turn contains a smaller circle with an 'X' inside. To the right of this is a sketch of a function curve labeled f and S , with a vertical line and a small circle below it.
- Bottom left: A series of five small circles arranged in a horizontal line.
- Bottom center: A large circle containing a five-petaled flower-like shape.
- Bottom right: A large circle containing a square, which in turn contains a smaller circle with an 'X' inside.

$$\int \frac{f(z)}{g(z)} dz \quad \leftrightarrow \quad \frac{2\pi \cdot i \cdot \text{Res}}{1}$$